Data Science

	Normative 4-year Degree
Major	Data Science 數據科學
Degree / Award Title	Bachelor of Science in Data Science 理學士(數據科學)
Offering Academic Unit	Department of Data Science
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2401 English for Science	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12

College-specified Courses	CS1315 Introduction to Computer Programming	3
	SDSC2003 Human Contexts and Ethics in Data Science	3
	CS3402 Database Systems	3

College / School Requirements (8 credit units) **Catalogue Term : Semester A 2025/26**

Course Code	Course Title	Credit Units
MA1503	Linear Algebra with Applications	4
MA1508	Calculus	4

Major Requirements (64 / 67 credit units) **Catalogue Term : Semester A 2026/27**

Notes:

The catalogue term of the major requirement that students will follow will be the effective term of the declared/allocated major.

For normative 4-year degree students who will join the majors allocation exercise, the catalogue term of major requirement will be one year after admission.

For advanced standing students and 4-year degree students who already have a major at the time of admission, the catalogue term of major requirement will be the same as their admission term.

1. Core Courses

(43 credit units for students without stream)
(52 credit units for students with ACT stream)

Course Code	Course Title	Credit Units	Remarks
SDSC1001	Introduction to Data Science	2	
MA2508	Multi-variable Calculus	4	
MA2510	Probability and Statistics	3	
SDSC2001	Python for Data Science	4	
SDSC2002	Convex Optimization	3	
SDSC2004	Data Visualization	3	
SDSC2005	Introduction to Computational Social Science	3	
SDSC2102	Statistical Methods and Data Analysis	3	
CS2334	Data Structures for Data Science	3	
CS3273	Data Protection and System Security	3	
SDSC3001	Big Data: The Arts and Science of Scaling	3	ACT stream only
SDSC3006	Fundamentals of Machine Learning I	3	
SDSC3007	Advanced Statistics	3	

SDSC3025	Internship for Flagship Programme	3	ACT stream only
SDSC3026	International Professional Development	3	ACT stream only
SDSC4116	Data Science Capstone	6	

2. Elective Courses

(21 credit units for students without stream)
(15 credit units for students with ACT stream)

Must earn at least 12 credit units from B4 level

Course Code	Course Title	Credit Units	Remarks
GE2339	Smart City - a Systems Engineering Perspective	3	B2 level / Artificial Intelligence
IS2505	e-Business	3	B2 level / General
CS3201	Computer Networks	3	B3 level / General
CS3342	Software Design	3	B3 level / General
CS3383	AI for Software Engineering	3	B3 level / General
LT3233	Computational Linguistics	3	B3 level / General
SDSC3001	Big Data: The Arts and Science of Scaling	3	B3 level / Artificial Intelligence, Core course for ACT stream
SDSC3002	Data Mining	3	B3 level / Artificial Intelligence and Statistical Learning
SDSC3003	Blockchain	3	B3 level / General
SDSC3004	Computational Optimization	3	B3 level / Statistical Learning
SDSC3005	Computational Statistics	3	B3 level / Statistical Learning
SDSC3008	Systems Dynamics and Control	3	B3 level / Artificial Intelligence
SDSC3009	Behavioural Analytics	3	B3 level / Social Media Analytics
SDSC3010	Digital Trace Analytics	3	B3 level / Social Media Analytics
SDSC3011	Social Data Processing and Modelling	3	B3 level / Social Media Analytics
SDSC3013	Introduction to Social Media Analytics	3	B3 level / Social Media Analytics
SDSC3014	Introduction to Sharing Economy	3	B3 level / General
SDSC3015	Knowledge Graph and Cognitive Computing	3	B3 level / Artificial Intelligence
SDSC3016	Social Network Analysis	3	B3 level / Social Media Analytics
SDSC3017	Game Theory and Its Application	3	B3 level / Artificial Intelligence

SDSC3018	Introduction to Internet of Things	3	B3 level / General
SDSC3019	Introduction to Networked Life and Data Science	3	B3 level / General
SDSC3022	Financial Data Analytics for Investments	3	B3 level / General
SDSC3023	Data Science Applications in Portfolio Risk Analysis	3	B3 level / General
SDSC3060	Operations Research	3	B3 level / General
SDSC3105	Bayesian Analysis	3	B3 level / Statistical Learning
COM4511	Social Media and Communication	3	B4 level / Social Media Analytics
CS4186	Computer Vision and Image Processing	3	B4 level / General
CS4286	Internet Security and e-Commerce Protocols	3	B4 level / General
CS4296	Cloud Computing	3	B4 level / General
CS4480	Data-Intensive Computing	3	B4 level / General
CS4486	Artificial Intelligence	3	B4 level / Artificial Intelligence
CS4487	Machine Learning	3	B4 level / Artificial Intelligence
SDSC4001	Foundation of Reinforcement Learning	3	B4 level / Artificial Intelligence and Statistical Learning
SDSC4008	Deep Learning	3	B4 level / Artificial Intelligence and Statistical Learning
SDSC4009	Data Intelligence in Action	3	B4 level / Artificial Intelligence
SDSC4010	Projects in Data Science (research)	3	B4 level / General
SDSC4011	Experimental Research for Social Media	3	B4 level / Social Media Analytics
SDSC4016	Fundamentals of Machine Learning II	3	B4 level / Artificial Intelligence, Statistical Learning and Social Media Analytics
SDSC4018	AI in Systematic Trading	3	B4 level / Artificial Intelligence
SDSC4019	Stochastic Processes and Applications	3	B4 level / Statistical Learning
SDSC4026	Systems Modelling and Simulation	3	B4 level / Social Media Analytics
SDSC4070	Large Language Models	3	B4 level / General
SDSC4110	Statistical Design and Analysis of Experiments	3	B4 level / Statistical Learning

3. Optional Electives for all Normative 4-year Degree

Students may choose to enrol in the below course(s) to enrich their study.

Course Code	Course Title	Credit Units	Remarks
SDSC0001	Internship	0	
SDSC0002	Internship	0	
SDSC0003	Internship	0	
SDSC0004	Internship	0	
SDSC3080	Internship	3	Not available for ACT stream